UI/UX Design Documentation

Project: Legal Document RAG Assistant

Inspiration Reference: "AI Business Builder Website" (Dribbble Shot #27224655 by Pam Sykes / Orizon)

1. Design Philosophy

The goal is to elevate a standard internal enterprise tool (a RAG legal assistant) into a premium, cinematic, and highly intuitive experience. The design will leverage a dark mode "cave-moonlight" atmosphere that feels sophisticated and focused. It will utilize floating elements and a clean, prompt-based interface to make complex legal queries feel effortless across all devices.

2. Color Palette

Based on the reference design, the application will use a deeply saturated, dark-green monochromatic theme with bright neon accents to indicate AI processing and system status.

- **Primary Background:** `#09100B` (Deep Dark Green/Black) - Used for the main app background to create the "cinematic" depth.
- **Secondary Background / Panels:** `#265131` (Forest Green) - Used for floating cards, chat bubbles, and settings panels.
- **Primary Text:** `#E9F4E2` (Pale Mint/Off-White) - High contrast for maximum readability of dense legal texts.
- **Secondary Text / Muted Elements:** `#919491` & `#5F655F` (Muted Gray-Greens) - Used for timestamps, metadata, and placeholder text.
- **AI Accents & Primary Buttons:** `#51C445` (Bright Neon Green) - Used for the "Submit Query" button, AI loading indicators, and highlighting retrieved legal entities.
- **Secondary Accents:** `#6AB597` & `#9AE5C8` (Teal/Light Teal) - Used for borders, hover states, and data visualizations (like Chroma similarity scores).

3. Typography

- **Primary Font:** A clean, modern sans-serif like *Inter*, *Satoshi*, or *Geist*.
- **Hierarchy & Responsive Scaling:**
  - **Headers:** Semibold, bright white/pale mint. (Scales down on mobile to prevent text wrapping issues).

- **Body Copy:** Regular weight, optimized for reading long legal clauses. Base size of 16px to prevent automatic zooming on mobile iOS devices when inputting text.
- **Code/System Output:** A monospaced font (like *JetBrains Mono*) for displaying similarity thresholds or metadata.

4. Core UI Components & Layouts

4.1 The "Hero" & Chat Interface (The Prompt UI)

- **Layout:** A centered, prompt-based UI inspired by the reference's clean layout. The chat window takes center stage, floating above the `#09100B` background.
- **Input Box:** A softly rounded rectangular input field anchored at the bottom of the screen. It features a glowing `#51C445` border when focused, signaling that the AI is ready. On mobile, this component uses `dvh` (dynamic viewport height) to ensure it stays above the virtual keyboard.
- **Chat Bubbles:**
  - *User Queries:* Aligned right, subtle `#265131` background.
  - *AI Responses:* Aligned left, no background (or very faint glassmorphism effect), emphasizing the text.

4.2 Context Citations (The "Social Proof" translation)

- **Concept:** The reference design features floating "social proof" cards. In our RAG app, these display the **Retrieved Document Chunks**.
- **PC Execution (Hover):** Subtle glassmorphic cards float to the right of the main chat. Hovering over a card highlights the specific section of the text.
- **Mobile Execution (Tap):** Since hover states do not exist on mobile, context citations are represented as inline superscript chips (e.g., `[1]`, `[2]`). Tapping a chip triggers a smooth "Bottom Sheet" modal containing the legal chunk.

4.3 The "Cave-Moonlight" Atmosphere (Visual Effects)

- **Background Effect:** A subtle, radial gradient originating from the top center or behind the main chat interface, transitioning from `#265131` to the pitch-black `#09100B`. This creates the "moonlight" effect, drawing the user's eye to the center prompt.

5. Mapping RAG Features to the UI

RAG Feature	UI/UX Execution

PDF Ingestion / Upload	A minimal, dashed dropzone in the center of the screen on the first load. On mobile, a prominent "Tap to Upload" button replaces the dropzone. A smooth glowing progress bar (`#9AE5C8`) indicates chunking.
Out-of-Context Guardrail	If the user asks an irrelevant question and the similarity threshold is triggered, the AI text color shifts to a muted `#919491` and types out: *"Sorry I can't find anything relevant to this question."* No context citations appear.
Chroma Similarity Scores	(For debugging/grading) A small toggle in the top right corner. When active, tiny neon green tags (`#51C445`) appear next to context citations displaying the Cosine Distance score (e.g., `Dist: 0.24`).

6. Universal Responsive & Cross-Device Strategy

To ensure a seamless experience without bugs on both PC and Mobile, the following structural rules apply:

6.1 Universal Constraints

- **Touch Targets:** All interactive elements (buttons, citations, toggles) must have a minimum touch target size of 44x44px for fat-finger mobile interactions.
- **Safe Areas:** Implementation must utilize CSS safe-area-insets to prevent the UI from being obstructed by mobile device notches or home indicators.
- **No Horizontal Scroll:** `overflow-x: hidden` applied globally to prevent layout breakage from long text strings or URLs.

6.2 PC / Desktop View (> 1024px)

- **Layout:** Fluid 3-column layout.
  - Left Sidebar (20%): Chat history.
  - Center Column (50%): Main prompt UI and chat.
  - Right Sidebar (30%): Floating legal context cards.
- **Interactions:** Relies on mouse-hover states for highlighting text and expanding context cards. Input box is anchored to the bottom.

6.3 Tablet View (768px - 1024px)

- **Layout:** 2-column layout.
  - Left sidebar hides behind a hamburger menu.
  - Center chat takes up 65% of the screen.
  - Right context sidebar shrinks to 35%.

6.4 Mobile View (< 768px)

- **Layout:** 1-column, full-screen chat interface.
- **Navigation:** Chat history and settings are moved entirely into an off-canvas hamburger menu.
- **Virtual Keyboard Protection:** The chat container must use `height: 100dvh` (Dynamic Viewport Height) so the input box perfectly rests above the mobile device's virtual keyboard when activated, rather than being covered by it.
- **Context Modals:** The right sidebar is completely removed. Legal citations become tappable inline links that pull up a half-screen "Bottom Sheet" overlay with a drag-down-to-dismiss feature.